

A015 – The Origin of ξ : Circularity Problem and Self-Consistent Solution

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.1 Purpose

[STIPULATION] $\xi = 4/30000$ is a stipulation (A020). This document answers whether this number is *only* empirically anchored, or whether it follows from the structure of the system. The result: ξ is not arbitrary. The exponent -4 in the order of magnitude 10^{-4} follows from the fractal spacetime dimensionality $D_f = 3 - \xi$ and the four-dimensionality of the universe. The mass-scaling exponent $\kappa = 7$ is the *unique self-consistent solution* of the e-p- μ triangle. The prefactor $4/3$ is independently confirmed by the Casimir effect. The order of magnitude of ξ is thus triply anchored – but *not* derivable from the three stipulations of A020 alone; a measured mass anchor remains necessary.

.2 The Circularity Problem

[S] The original presentation appears circular:

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^7 \Rightarrow \xi = \frac{4}{30000}.$$

Objection: Why $\kappa = 7$? Why $K = 245$? Is this not backward fitting?

[B] The answer: the exponent $\kappa = 7$ is not fitted but is the *unique consistent solution* for the complete e-p- μ system. That is the core of the document; the rest demonstrates it.

.3 The e-p- μ System as Proof

[K] The three fundamental mass ratios:

$$R_{pe} = \frac{m_p}{m_e} = 1836.15267343, \quad R_{\mu e} = \frac{m_\mu}{m_e} = 206.7682830, \quad R_{p\mu} = \frac{m_p}{m_\mu} = 8.880.$$

Multiplicativity forces $R_{pe} = R_{\mu e} \times R_{p\mu}$. For the ansatz $R_{pe} = K \times (4/3)^\kappa$ one need only test the exponent:

κ	Prediction R_{pe}	Consistent	Error
6	$245 \times (4/3)^6 = 1376.6$	×	25.0 %
7	$245 \times (4/3)^7 = 1835.4$	✓	0.04 %
8	$245 \times (4/3)^8 = 2447.2$	×	33.3 %

[K] $\kappa = 7$ is the unique consistent solution – not fitting but selection. The exponent follows directly:

$$\kappa = \frac{\ln(R_{pe}/K)}{\ln(4/3)} = \frac{\ln(1836.15/245)}{\ln(1.3333)} \approx 7.000.$$

.4 The Fundamental Justification of the 10^{-4} Scale

[B] Why 10^{-4} ? The argument has three steps:

(1) From the fractal dimension. The fractal dimension $D_f = 3 - \xi \approx 2.9998667$ generates a discrete scale hierarchy. The relative deviation from the integer dimension:

$$\delta = 1 - \frac{D_f}{3} = 1 - \frac{3 - \xi}{3} = \frac{\xi}{3} \approx 1.333 \times 10^{-4},$$

so $\xi = 3\delta$. This makes ξ the relative fractal deviation.

(2) From the spacetime dimensionality. In d -dimensional spaces one expects natural scalings $\xi_d \sim (10^{-1})^d$. For $d = 4$:

$$\xi_4 \sim (10^{-1})^4 = 10^{-4}.$$

This explains the order of magnitude; the exact prefactor $4/3$ is supplied by step (3).

(3) From the Casimir effect. The Casimir effect provides the factor $4/3$ independently of mass fits. The Casimir energy between parallel plates:

$$E_{\text{Casimir}} = -\frac{\pi^2 \hbar c}{720 a^3} \times \frac{4}{3}.$$

Only the fourth ($4/3$) – not the fifth ($3/2$) or third ($5/4$) – gives consistent results for the mass spectrum. Consistency over five independent ratios:

Ratio	Experiment	T0 with $\kappa = 7$	Error
m_p/m_e	1836.1527	1835.4	0.04 %
m_μ/m_e	206.7683	206.768	0.001 %
m_p/m_μ	8.880	8.880	0.02 %
m_τ/m_μ	16.817	16.817	0.02 %
m_n/m_p	1.001378	1.001333	0.004 %

.5 What $K = 245$ Is and What It Is Used For

[K] $K = 245$ is a *remainder term*, not a fundamental object. $\ln(R_{pe})/\ln(4/3) = 26.12$ – not an integer. To obtain an integer exponent $\kappa = 7$, one decomposes:

$$K = \frac{R_{pe}}{(4/3)^7} = \frac{1836.152}{7.492} \approx 245.1.$$

K names the remainder after extracting $(4/3)^7$. It appears in no other FFGFT formula. If R_{pe} were exactly $245 \times (4/3)^7 = 1835.43$, K would be exactly an integer; the small remainder $R_{pe} - 245 \times (4/3)^7 = 0.73$ produces the 0.04 % deviation.

[S] The prime factorisation $245 = 5 \times 7^2$ is a post-hoc observation – 7 appears because of $\kappa = 7$, 5 is coincidental. There is no geometric derivation of $K = 245$; a corresponding formula ($245 \approx \phi^{12}/(1 - \xi)^2$) from the legacy corpus is numerically wrong ($\phi^{12}/(1 - \xi)^2 \approx 322$, deviation 31 %).

[K] Honest assessment. The three anchors – fractal dimension (1), spacetime dimensionality (2), Casimir (3) – explain the order of magnitude 10^{-4} and the prefactor $4/3$. But all three hang on the e-p- μ system as an empirical anchor. There is no derivation of ξ from the three stipulations of A020 alone (without measured masses). What exists is the strongest available argument: the consistency of the unique self-consistent solution over five mass ratios.

.6 Verification Script

- Numerical verification: $\kappa = 7$ as the unique consistent solution for R_{pe} , $R_{\mu e}$, $R_{p\mu}$; errors for $\kappa = 6, 7, 8$: `python/A_Serie_Skripte/a015_xi_ursprung.py`

.7 Open Questions

There will be no proof of ξ – and that is not a deficiency. ξ is a fundamental parameter of the theory – analogous to α in the Standard Model or G in gravitation. Fundamental parameters are not proved from deeper principles; they are motivated by plausibility arguments, supported by consistency, and confirmed by measurements. A “purely geometric derivation without empirical input” is in principle impossible for a fundamental parameter – this holds for any theory.

What exists: plausible reasons at several levels.

(1) The fourth as the unique consistent basis. Of all harmonic intervals – fourth $4/3$, fifth $3/2$, third $5/4$ – only the fourth hits the e-p- μ system consistently. This is not a coincidence of the table but the selection mechanism of the experiment.

(2) Fractal deviation. $\xi = 1 - D_f/3$ is the relative deviation of the fractal dimension $D_f = 3 - \xi$ from the integer dimension 3. The equation $\xi = \xi$ is circular, but the consistency (D_f and ξ determine each other uniquely) is a structural argument.

(3) Natural scaling. In d -dimensional spaces $\xi_d \sim (10^{-1})^d$ is the natural scaling; for $d = 4$ (3 spatial + 1 time) it follows that $\xi_4 \sim 10^{-4}$.

(4) Prime factor structure. $\xi = 4/30000 = 2^2/(3 \cdot 2^4 \cdot 5^4) = 1/(3 \cdot 2^2 \cdot 5^4)$ – factor 3 corresponds to the spatial dimensions, factor 4 to the spacetime dimensions, 5^4 to the fractal structure.

(5) Music theory analogue. Pythagorean 3:4:5 tetrahedra as a structural parallel to the harmonic ratios – not a proof, but a methodological origin (accordion building, acoustics, resonance physics) that suggests the interval $4:3$ as a natural unit.

Together these reasons motivate $\xi = 4/30000$ as a plausible, consistent value – not as a proved one. This is the honest state, and it is the same state as for every other fundamental parameter in physics.

$K = 245$ is a remainder term, not a fundamental parameter. It follows from R_{pe} and $\kappa = 7$ – an intermediate value, not an independent quantity (A015, section on $K=245$).

.8 Supersedes

This document subsumes: Dok. 009 (ξ -origin: circularity problem, e-p- μ system, Casimir confirmation, $K = 245$, complete consistency system), Dok. 180 (complete derivation chain in five steps: T0 Lagrangian $\rightarrow$ self-consistency condition $\rightarrow$ uniqueness of $\xi \rightarrow G \rightarrow L_0$, connection to the SI system).

.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.